



BATTERY ENERGY STORAGE SYSTEM



BESS SYSTEM MAP

1. Battery Containers

2. PCS/MV skids

3. MV combiner switchgear

4. HV switchyard

5. Main step-up transformer

6. Grid / POI

7. Auxiliary supply

8. Control room



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evault

ABOUT EVAULT

Enerveda Vault is an Indian manufacturer of utility-scale Battery Energy Storage Systems (BESS), built to meet the evolving demands of India's power infrastructure. From our advanced manufacturing facility in Gujarat, we design and assemble liquid-cooled, containerized LFP battery systems in 261 KWH , 418KWH , 1044 KWH - 5 MWh and 6.25 MWh DC block configurations.

Every system is engineered, tested, and certified by India's leading engineering talent to deliver reliable, safe, and scalable energy storage for commercial, industrial, utility, and renewable energy applications across India and global markets.

**Built on engineering excellence,
backed by measurable capabilities.**

1,00,000 sq ft

Manufacturing facility Bavla, Gujarat

5 GWh

Annual production capacity

80 units/month

Assembly, integration and testing

8,000+ cycles

Design life across all products

Every Unit. Tested. Logged. Proven.

Six facts about how we build.



01 Four-level traceability

Every cell is validated and logged across four stages—cell, module, pack, and final BESS factory acceptance test. All units moves forward only after certification.



02 Full-power tested Every one

Every container goes through a complete charge-discharge cycle at rated power before dispatch.



04 Factory compatible to large format cells

Our production line is engineered to adopt next generation, large format cells including 588ah and 628ah.



03 Laser-welded cell joints

Precision IPG fibre laser welding delivers consistent, low-resistance joints across every cell interconnection



05 Rain-tested at rated IP class

Every container is validated against water ingress before it ships out of the factory. It is designed and tested as per IP55 rating.



06 Certification-led manufacturing

All products are IEC and UL certified following the highest standards of safety and performance.

**We power
our own factory
with our own batteries.**



What We Build

Two systems. One engineering standard.



TITAN

Grid-scale energy storage for utility infrastructure.

* **TITAN - 5 MWh DC Block**

For the infrastructure that holds the country together.

Grid operator

State DISCOMs

SECI

Renewable integration

CORE

Liquid-cooled energy storage for commercial and industrial operations.

For businesses where power is not a utility. It is an operating condition.

- * **CORE 261**
- * **CORE 418**
- * **CORE 1044**

Data centres

Manufacturing

Hotels

Re Parks

Hospitals



261 kWh / 125 kW

CORE



Liquid-Cooled Energy Storage System

Reliable power for businesses that run around the clock. Liquid-cooled thermal management keeps cell temperature consistent within ± 2.5 degC across every cycle - eliminating the hot spots that accelerate degradation in air-cooled systems, extending cycle life, and removing the thermal runaway risk that air cooling cannot fully contain. Modular, expandable, and remotely monitored. A plant that installs CORE 261 today can expand capacity without replacing the system it already owns

DC Side	Cell Type	LFP / 314 Ah
	Pack Configuration	52.2 kWh / 1P52S
	System Configuration	261 kWh / 1P260S
	Rated DC Voltage	832 V
	DC Voltage Range	728 - 936 V
	Max. Charge/Discharge Rate	0.5 P
	Max. Depth of Discharge	100% (25 ± 2°C)
AC Side	Rated Output Power	125 kW
	Rated AC Voltage	400 V
	AC Voltage Range	±15%
	Grid Type	3W + N + PE
	Rated Frequency	50 Hz / 60 Hz
	Power Factor	0.99 / -1+1
	THDi	≤ 3%
General	DC Ratio	< 0.5% I _{pn}
	Max. Round Trip Efficiency	89 %
	Cycle Life	≥ 8,000 cycles
	Communication	Modbus TCP/IP
	Fire Suppression System	Aerosol
	Ingress Rating	IP55
	Cooling	Liquid cooling + Forced air cooling
	Operating Temperature	-25°C -55°C (Derating after 45°C)
	Anticorrosion Rating	C4 (C5 optional)
	Humidity	0 - 95% RH (non-condensing)
	Noise	≤ 75 dB
	Altitude	3000m (Derating above 2000m)
	Dimensions (W*D*H)	1,050 * 1,350 * 2,400 mm
	Weight	2,600 kg

418 kWh

CORE



Liquid-Cooled Energy Storage System

Higher capacity on the same engineering platform. Designed for larger commercial and industrial loads - data centres running at scale, manufacturing plants with continuous shift operations, hospital campuses where stored energy is also emergency infrastructure. At this capacity, the financial case is direct: the stored energy pays for the system within a predictable number of billing cycles. The decision is not whether to invest. It is when.

		418 - 40	418 - 69	418 - 80
DC Side	Cell Type	LFP / 314 Ah		
	Pack Configuration	52.248 kWh / 1P52S		
	System Configuration	418 kWh / 1P416S		
	Rated DC Voltage	1331.2 V		
	DC Voltage Range	1164.8 ~ 1497.6 V		
	Max. Charge/Discharge Rate	0.25 P	0.5 P	0.5 P
	Max. Depth of Discharge	100% (25 ± 2°C)		
AC Side	Rated Output Power	100 kW	215 kW	210 kW
	Rated AC Voltage	400 V	690 V	800 V
	AC Voltage Range	-15% ~ +10%		
	Grid Type	3W + PE		
	Rated Frequency	50 Hz / 60 Hz		
	Power Factor	0.99 / -1 ~ +1		
	THDi	≤ 3%		
	DC Ratio	< 0.5% I _{pn}		
General	Max. Round Trip Efficiency	89%		
	Cycle Life	≥ 8,000 cycles		
	Communication	Modbus TCP/IP		
	Fire Suppression System	Aerosol		
	Ingress Rating	IP55		
	Cooling	Liquid cooling + Forced air cooling		
	Operating Temperature	25°C ~ 55°C (Derating after 45°C)		
	Anticorrosion Rating	C4 (C5 optional)		
	Humidity	0 ~ 95% RH (non-condensing)		
	Noise	≤ 80 dB		
	Altitude	3000 m (Derating above 2000 m)		
	Dimensions (W*D*H)	1,500*1,350*2,400 mm		
	Weight	3,800 kg		

1044

CORE



Liquid-Cooled Energy Storage System

A highly integrated, standard 10-foot containerized energy storage system designed for commercial and industrial applications, with a rated AC output voltage of 400V. This system is capable of on/off-grid switch functionality, and the STS and <20ms seamless switching feature is available as an optional upgrade, providing strong off-grid load capability. It is equipped with dedicated interfaces for diesel generator connection and PV inverter connection, enabling multiple power sources to supply the system. This flexible design allows the system to adapt to various application scenarios and diverse user requirements.

TECHNICAL PARAMETERS

	CellType	LFP/314 Ah
	Pack Configuration	52.2 kWh/1P52S
	System Configuration	1044 kWh/4P260S
	Rated DC Voltage	832
	DC Voltage Range	728-936 V
	Max. Charge/Discharge Rate	0.5 P
	Max. Depth of Discharge	100% (25±2°C)
POWER SOURCE	Rated ESS Power	500 kW
	Rated AC Voltage	400 V
	AC Voltage Range	±15%
	Grid Type	3W+N+PE
	Rated Frequency	50 Hz/60 Hz
	Rated STS Power	1000kW
	On/Off-grid Switching	<20ms
	Power Factor	0.99/-1+1
	THDI	<0.5% Ipn
	Max. PV Inverter Input Current	800 A
	Max. Diesel Generator Input Current	800 A
BACKUP	Max. Backup Load Port Current	800 A
	Max. Backup Load Power*	350 kW (70% of Rated Output Power)
GENERAL	Cycle Life	≥ 8,000 cycles
	Communication	Modbus TCP/IP
	Fire Suppression System	Aerosol system+Water spray system
	Ingress Rating	IP55
	Cooling	Liquid cooling+Forced air cooling
	Operating Temperature	-25°C-55°C (Derating after 45°C)
	Anticorrosion Rating	C4 (C5 optional)
	Humidity	0-95% RH (non-condensing)
	Noise	≤ 80 dB
	Altitude	3000m (Derating above 2000m)
	Dimensions (W*D*H)	2991 * 2438 2896 mm
	Weight	14 t

5 MWh DC Block

TITAN



Grid-Scale Energy Storage

the 20 ft liquid cool ESS container integrates pack EMS, BMS,HVAC and fire suppression system (FFS) into a single container designed for demanding applications the, 20 ft liquid cool ESS container is suitable for power generation, grid and commercial and industrial (C&I) ESS scenarios that require high power and flexible capacity.

High Safety - Multi-layer fire protection

PACK-level and cabin-level gas fire suppression, cabin-level water extinguishing, and aerosol fire suppression for electrical cabinets. Not one system. Three.

Easy O&M - Front-access design

Front-access maintenance, quick module replacement, intelligent control management. Reduced operational costs over the 10-year service life - because the total cost of ownership matters more than the purchase price.

High Energy Density

Compact 20-foot container design with precise liquid cooling for long life. 5 MWh in a standard footprint - deployable at DISCOM substations without additional civil works.

High Intelligence - Integrated BMS/PCS/EMS

Integrated BMS, PCS, and EMS data supporting real-time failure diagnosis and prediction. The system knows before the operator knows.

Battery System	Cell type	LFP 3.2V / 314 Ah
	Battery capacity	5 MWh DC Block
	Rated Voltage	1,331.2 V
	C-Rate	<= 0.5P
	Discharge Efficiency	94%
	DoD	90% (adjustable 0-100%)
	Cycle Life	>= 8,000 cycles
Protection & Environment	IP Rating	IP55 (Pack IP67)
	Operating Temperature Range	Charge: 0 deg C to 55 deg C Discharge: -20 deg C to 55 deg C
	Fire Protection System	Fire aerosol + water spray (pack and container-level)
	Cooling Method	Intelligent liquid cooling
	Altitude	3,000 m (derating above 2,000 m)
Communication & Compliance	Communication Protocols	CAN / RS485 / LAN
	Compliance	IEC 62933-5-1, IEC 62933-5-2, UL 9540A, IEC 62619
	Weight	~ 43 T
	Dimensions	6,058 x 2,438 x 2,896 mm (20-ft container)



Service & Support that delivers confidence from project design to post-installation, Evaults provides a full-service energy storage experience.



Solution Design

Application-based ESS sizing, system planning & simulations



Pre-Sales Engineering

Technical documentation, drawings & remote commissioning guidance



After-Sales Service

Spare parts, warranty, inspections, and local partner network



Local Support

Global service network with regional engineers and partners



On-Site Delivery & Startup

Fast deployment with pre-assembled systems



24/7 Remote Monitoring

Full alarms, performance analytics, and firmware upgrades



Predictive Maintenance

AI-powered analytics for fault prevention and efficiency



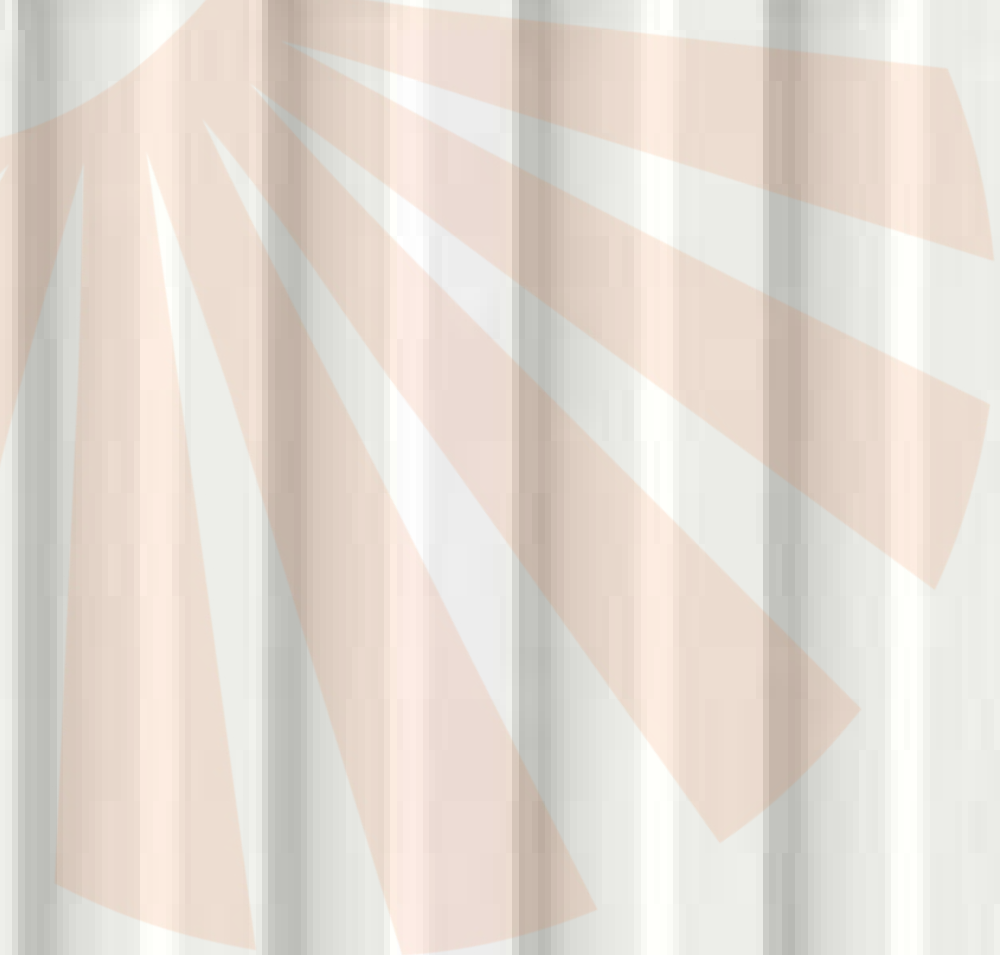
Spare Parts Readiness

Fast-response parts centers in key overseas markets



Visualized Service

Cloud-based platform with full lifecycle tracking & reports



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